

Ainesh Sabharwal

832-302-6497 | aineshsab@berkeley.edu | [linkedin.com/in/ainesh-sabharwal](https://www.linkedin.com/in/ainesh-sabharwal) | github.com/sabharwalainesh

EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science

Berkeley, CA

Aug. 2026 – May 2029

Bridgeland High School

High School Diploma, Graduated Salutatorian (9/921 people)

Cypress, TX

Aug. 2022 – May 2026

EXPERIENCE

Undergraduate Research Assistant

University of California, Berkeley

September 2026 – Present

Berkeley, CA

- Building a Gemini extraction pipeline with structured JSON outputs to classify municipal regulations from LOCUS's 2.2M-row corpus
- Developing Python and DuckDB filters to recover 168K overlooked licensing and permitting sections
- Creating a multi-annotator review interface to validate model outputs against human annotations

Lead Instructor

iCode School Franchise

Oct. 2025 – Mar. 2026

Cypress, TX

- Taught Python, Java, and Roblox development fundamentals through hands-on projects for K–12 students
- Partnered with the lead director to redesign standardized curriculum to match students' skill levels
- Guided students through coding projects, building debugging and problem-solving skills

Teaching Assistant

Self-Employed

Aug 2025 – May 2026

Cypress, TX

- Volunteered with the CS department to support grading for a high school Data Structures and Algorithms course
- Created instructional videos to explain algorithms and reinforce programming concepts
- Developed worksheets and practice exercises to strengthen students' understanding of data structures and problem-solving

PROJECTS

ApathyAI | *Python, Tauri, Ollama, LanceDB*

June 2026 – Present

- Co-developed a desktop AI copilot that interprets screen context and summarizes documents using local LLMs
- Integrated accessibility APIs, OCR, and vector search to retrieve relevant context from recent workflows
- Added API authentication, rate limiting, and sensitive-data redaction to protect captured screen content

Bridgeland Arena Hub | *TypeScript, React, Supabase, React Three Fiber*

Oct. 2024 – July 2025

- Co-developed an event ticketing platform that placed 3rd nationally in FBLA Website Design
- Built interactive 2D and 3D seating charts with seat selection and section-based pricing
- Integrated PayPal checkout, ticket history, and administrative tools for event management

Drawly | *Swift, SwiftUI, MultipeerConnectivity, PencilKit*

July. 2026 – Current

- Developing an iOS and macOS note-taking app with real-time, peer-to-peer synchronization
- Implementing shared Swift modules for note editing, device communication, and media attachments
- Integrating document scanning, speech transcription, and sketch-to-SVG export using Apple frameworks

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Swift, HTML/CSS

Frameworks: React, SwiftUI, Tauri, Tailwind CSS

Developer Tools: Git, GitHub, PostgreSQL, Supabase, DuckDB, Vite, Vercel

Libraries: Gemini API, Ollama, LanceDB, Google Places API